S3 TOC Class Notes on 03/08/2026, Monday, 11.30 am to 12.30 pm

Formal Definition of NFA

NFA (Non-deterministic Finite Automaton)

An NFA, denoted by **M**, is a **5-tuple**:

$$M = (Q,\ \Sigma,\ \delta,\ q_0,\ F)$$

where:

- **Q** is a finite, non-empty set of states.
- **Σ** is a finite, non-empty set of input symbols (input alphabet).
- **δ** is the transition function, which maps

$$\delta : Q \times \Sigma \to 2^Q$$

- **q_0** is the initial state, where

$$q_0 \in Q$$

- **F** is the finite, non-empty set of final (accepting) states, where

$$F \subseteq Q$$

Note: The notation 2^Q denotes the **power set of** Q, i.e., the set of all subsets of Q. This is because an NFA can transition to **zero, one, or more states** for a given input symbol.

Language Acceptability of NFA

Any language accepted by a **DFA** can also be accepted by an **NFA**, and vice versa. Hence, **DFA is a special case of NFA** in which there is **exactly one transition** for each input symbol from every state.

Consider an NFA **M** with current state **q** and input symbol **a**.

- If **δ(q, a)** is a set containing **n** states, then NFA **M** splits into **n identical copies** of itself.
- Each copy follows one of the possible transitions in **δ(q, a)**.
- This process is called **parallel computation**.
- If a copy encounters

$$\delta(q, a) = \varnothing$$

  then that copy **dies**, while the remaining copies continue their computation.
- If **at least one copy** of the NFA reaches a **final state** after successfully processing the input string **x**, then the NFA **accepts** the string **x**.
- Otherwise, the string is **rejected**.

Hence, the language accepted by an NFA is defined as

$$L(M) = \{ x \mid \delta(q_0, x) \cap F \neq \varnothing \}$$

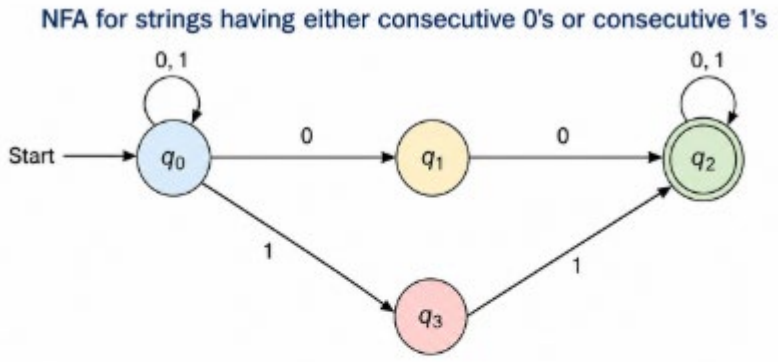
where:

- q_0 is the initial state,
- F is the set of final states,
- $\delta(q_0, x)$ is the set of states reachable after processing the string x.

Example 2

Design an NFA for strings having either consecutive 0's or consecutive 1's. Trace the input string `01001`.

NFA



$Q = \{ q_0, q_1, q_2, q_3 \}$
$\Sigma = \{ 0, 1 \}$
Start state = q_0
Final state = $\{ q_2 \}$

Transition function $\delta : Q \times \Sigma \rightarrow 2^Q$

State	δ(state, 0)	δ(state, 1)
q_0	$\{ q_0, q_1 \}$	$\{ q_0, q_3 \}$
q_1	$\{ q_2 \}$	$\varnothing$
q_3	$\varnothing$	$\{ q_2 \}$
q_2	$\{ q_2 \}$	$\{ q_2 \}$

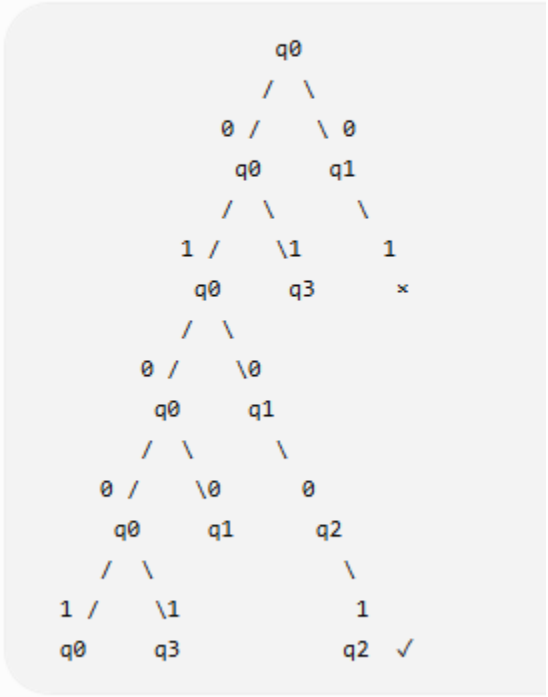
Transitions:

- $\delta(q_0, 0) = \{q_0, q_1\}$
- $\delta(q_0, 1) = \{q_0, q_3\}$
- $\delta(q_1, 0) = \{q_2\}$
- $\delta(q_3, 1) = \{q_2\}$
- $\delta(q_2, 0) = \{q_2\}$
- $\delta(q_2, 1) = \{q_2\}$

Final state: q_2

Trace for Input String: `01001`

Computation Tree



(`×` denotes a dead path.)

Accepting Sequence

```
q0 → q0 → q0 → q1 → q2 → q2
```

Since at least one computation reaches the final state $\mathbf{q_2}$, the input string `01001` **is accepted** by the NFA.

Computation of $\delta(q_0, 01001)$

Using the extended transition function,

$$\begin{aligned}
\delta(q_0, 01001) &= \delta(\delta(q_0, 0), 1001) \\
&= \delta(\{q_0, q_1\}, 1001) \\
&= \delta(q_0, 1001) \cup \delta(q_1, 1001) \\
\\
&= \delta(\delta(q_0, 1), 001) \cup \delta(\delta(q_1, 1), 001) \\
&= \delta(\{q_0, q_3\}, 001) \cup \delta(\phi, 001) \\
\\
&= \delta(\delta(q_0, 0), 01) \cup \delta(\delta(q_3, 0), 01) \\
&= \delta(\{q_0, q_1\}, 01) \cup \delta(\phi, 01) \\
\\
&= \delta(\delta(q_0, 0), 1) \cup \delta(\delta(q_1, 0), 1) \\
&= \delta(\{q_0, q_1\}, 1) \cup \delta(\{q_2\}, 1)
\end{aligned}$$

$$\begin{aligned}
&= \delta(q_0, 1) \cup \delta(q_1, 1) \cup \delta(q_2, 1) \\
&= \{q_0, q_3\} \cup \phi \cup \{q_2\} \\
\\
&= \{q_0, q_2, q_3\}
\end{aligned}$$

Since

$$\delta(q_0, 01001) \cap F = \{q_0, q_2, q_3\} \cap \{q_2\} = \{q_2\} \neq \phi,$$

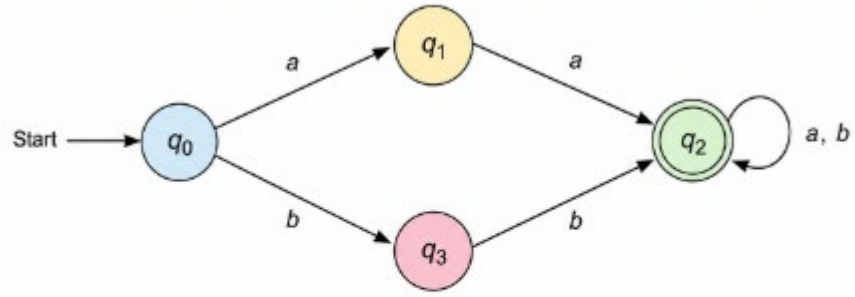
the string **01001 is accepted** by the NFA.

Example 3

Design an NFA for strings in which the first and second letters match.

$$\Sigma = \{a, b\}$$

NFA for strings in which first and second letters match



$M = (\{q_0, q_1, q_2, q_3\}, \{a, b\}, \delta, q_0, \{q_2\})$

States $(Q) = \{ q_0, q_1, q_2, q_3 \}$

Input alphabet $(\Sigma) = \{ a, b \}$

Start state $= q_0$

...al states $(F) = \{ q_2 \}$

Transition function δ

State	δ(state, a)	δ(state, b)
q_0	$\{ q_1 \}$	$\{ q_3 \}$
q_1	$\{ q_2 \}$	$\emptyset$
q_3	$\emptyset$	$\{ q_2 \}$
q_2	$\{ q_2 \}$	$\{ q_2 \}$

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